

Youngwoong Cho

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EDUCATION

- **Georgia Institute of Technology** Expected May 2028
M.S. in Robotics Atlanta, GA
- **The Cooper Union for the Advancement of Science and Art** Class of 2023
B.E. in Mechanical Engineering; Minor in Computer Science New York, NY
 - **Summa Cum Laude** | GPA: **3.92/4.00** | Engineering GPA: **3.96/4.00**
 - Norman Perry Internship Fund | Half Tuition Scholarship | Innovator's Merit Award | William C. and Esther Hoffman Beller Prize

RESEARCH EXPERIENCE

- **Robot Learning and Reasoning Lab (RL2), Georgia Tech** August 2026 – Present
Graduate Researcher | Prof. Danfei Xu Atlanta, GA
 - **Cross-Embodiment Dexterous Manipulation:** Studying whether co-training on more hand types improves zero-shot transfer to unseen hands, on a leave-one-hand-out benchmark of 7 simulated hands
 - **Learnability of Retargeted Demonstrations:** Studying which properties of retargeted action data (e.g., consistency, smoothness, diversity) make demonstrations easier for imitation learning on a real robot
 - **Cross-Embodiment Data Infrastructure:** Built an Apple Vision Pro–Isaac Sim teleoperation pipeline and a shared wrist–fingertip action interface that lets one policy control 7 dexterous hands
- **RLWRLD (Robotics Foundation-Model Startup)** April 2026 – August 2026
Research Engineer Seoul, Republic of Korea
 - **Fine-tuned GR00T N1.6 on DexJoCo**, an 11-task dexterous-hand benchmark, reaching **46.2%** mean success vs. 40.3% reported for N1.5; also evaluated released $\pi_{0.5}$ checkpoints
 - **Improved robot-state tokenization in the GR00T N1.6 VLA**, raising humanoid simulation success from 36.4% to **41.5%** by encoding state as 11 tokens instead of 1
 - **Adapted a 3D geometry-aware policy to a 16-DoF dexterous hand**, fixing a step-counting bug in its trainer and raising success on a single-arm task from 11% to **~55%**
 - **Built a training/evaluation platform on 3 compute clusters** and made MuJoCo evaluation bit-reproducible, eliminating run-to-run outcome changes in **22.5%** of same-seed episodes
- **Naver Labs** March 2022 – August 2022
Deep Learning Research Intern, Autonomous Driving | Siyeong Lee Seongnam, Republic of Korea
 - **Improved 3D bounding-box prediction** with a self-supervised iterative refinement module integrated into MuZero, increasing task success from 10% to **30%**
 - **Identified RL exploration inefficiency** and redirected the learning framework toward GAIL-based imitation learning inspired by IQ-Learn
 - **Built a 3D annotation tool** for collecting human demonstrations, reducing annotation time by **70%**
- **Seoul Robotics** May 2021 – March 2022
Machine Learning Research Engineer Intern | Dr. Hong Minh Truong Seoul, Republic of Korea
 - **Developed vectorized curb detection** for Seoul-specific driving environments, establishing the experimental baseline for subsequent drivable-area research
 - **Built a multi-view LiDAR annotation system** with temporal stacking and synchronized editing, producing labels that improved drivable-area detection by **15%**
 - **Standardized perception evaluation** across mAP, MOTA, and velocity metrics over **6,000+ frames**, producing the company's primary evaluation tool
- **Seoul National University** May 2020 – July 2020
Research Intern, Interactive & Network Robotics Lab | Prof. Dongjun Lee Seoul, Republic of Korea
 - **Validated cooperative manipulation algorithms** on a physical multi-robot testbed, identifying and debugging systematic sim-to-real gaps

INDUSTRY EXPERIENCE

- **Enmicht (Subsidiary of ESU Farm LLC)** March 2024 – December 2025
ERP Software Engineer Englewood Cliffs, NJ
 - **Built a closed-loop IoT monitoring system** integrating temperature, humidity, and 3-axis accelerometer sensors for real-time supply-chain quality control
 - **Implemented accelerometer sensor fusion** for state estimation, frequency-domain transport-anomaly detection, and impact-event classification, alongside thermal PID control
- **LG CNS** July 2023 – February 2024
Data Analyst, Customer Data Platform Englewood Cliffs, NJ
 - **Designed a unified analytics warehouse** in GCP BigQuery integrating Google Analytics, CRM, and internal data, reducing data-model redundancy by **80%**

SELECTED PROJECTS

- **Self-Driving Delivery Robot** Spring 2023
Senior Capstone, The Cooper Union
 - **Built an end-to-end autonomous navigation stack** (3D LiDAR SLAM, localization, planning, control) for an indoor delivery robot, with localization latency under **100 ms**
- **BarlowZero: Sample-Efficient Model-Based Reinforcement Learning** Fall 2022
ECE-472: Deep Learning, The Cooper Union
 - **Developed a MuZero variant with Barlow Twins self-supervision**, cutting training batch size from 512 to **64** while performing comparably to EfficientZero on Atari Breakout
- **Cooper Union IGVC** Fall 2022 – Spring 2023
Intelligent Ground Vehicle Competition
 - **Developed LiDAR- and RGB-based localization, mapping, and path planning** in ROS, plus an OCR-based stop-sign classifier
 - **Placed 2nd in autonomous navigation and 3rd in vehicle design**

TECHNICAL SKILLS

- **Robot Learning:** VLAs (GR00T N1.5/N1.6, $\pi_{0.5}$), Diffusion Policy, HAT, HPT, Imitation Learning, RL
- **Robotics:** Dexterous Manipulation, Retargeting (FK/IK), Teleoperation, Mobile Manipulation, SLAM
- **ML / Simulation:** PyTorch, FSDP, CUDA, Isaac Sim, MuJoCo, Weights & Biases
- **Programming:** Python, C++, TypeScript, JavaScript
- **Infrastructure:** Slurm, Kubernetes, ROS, Git, GCP BigQuery

ACTIVITIES & MEMBERSHIPS

- **Tau Beta Pi**, Member November 2020 – Present
- **Korean Association of Cooper Union**, Vice President August 2020 – May 2021